

Quantum Annealing Algorithm for MaxCut

We outline the algorithmic approach for solving the MaxCut problem using quantum annealing. The method is based on mapping the classical MaxCut objective to a quantum Hamiltonian and evolving the system from an initial simple state to the ground state of the problem Hamiltonian.

Algorithm

1. **Qubit Assignment:** Place one qubit on each vertex of the graph. Each qubit can have a spin ± 1 (or equivalently, be in the computational basis states $|0\rangle$ or $|1\rangle$).
2. **Partitioning:** Conceptually divide the set of vertices into two subsets, A and B , where qubits in A correspond to spin-up ($+1$) and qubits in B correspond to spin-down (-1).
3. **Pauli-Z Mapping:** Define the eigenvectors of the Pauli-Z operator such that the eigenvector with positive eigenvalue corresponds to spin-up, and the one with negative eigenvalue corresponds to spin-down.
4. **Classical Cost Function:** Express the MaxCut objective as

$$C(\{s_i\}) = \sum_{(i,j) \in E} \frac{1 - s_i s_j}{2}, \quad (1)$$

where $s_i \in \{+1, -1\}$ represents the spin of qubit i .

5. **Equivalent Minimization:** Equivalently, the maximization of C can be written as the minimization of

$$H_{\text{classical}}(\{s_i\}) = \sum_{(i,j) \in E} \frac{-1 + s_i s_j}{2}. \quad (2)$$

6. **Problem Hamiltonian:** Promote the classical spins s_i to Pauli-Z operators Z_i , giving the quantum problem Hamiltonian

$$H_P = \sum_{(i,j) \in E} \frac{-1 + Z_i Z_j}{2}. \quad (3)$$

The ground state of H_P corresponds to the maximum cut of the graph, and its minimum expectation value encodes the optimal cut size.

7. **Initial Hamiltonian:** Choose a simple, easily preparable initial Hamiltonian, typically the transverse field

$$H_B = - \sum_i X_i, \quad (4)$$

whose ground state is the equal superposition

$$|\psi_0\rangle = \bigotimes_{i=1}^n \frac{|0\rangle_i + |1\rangle_i}{\sqrt{2}}. \quad (5)$$

Reason: To make crossing less likely

8. **Interpolating Hamiltonian:** Define a time-dependent Hamiltonian that interpolates between the initial and problem Hamiltonians:

$$H(t) = (1 - s(t))H_B + s(t)H_P, \quad s(0) = 0, \quad s(T) = 1, \quad (6)$$

where $s(t)$ is a smooth schedule function controlling the interpolation. Linear interpolation is commonly used:

$$s(t) = \frac{t}{T}. \quad (7)$$

9. **Time Evolution:** Simulate the evolution of the quantum state under $H(t)$ from $t = 0$ to $t = T$. In practice, this is done by discretizing the evolution into small time steps Δt :

$$U(T, 0) \approx \prod_{k=1}^M e^{-iH(s_k)\Delta t}, \quad s_k = \frac{k}{M}. \quad (8)$$

Here, M is the total number of discrete steps.

10. **Measurement and MaxCut Extraction:** After evolution, measure the final state in the computational basis. Compute the expectation value

$$E = \langle \psi(T) | H_P | \psi(T) \rangle. \quad (9)$$

The minimum of E corresponds to the ground-state energy of H_P , and its magnitude gives the size of the maximum cut. The bitstring(s) achieving this energy yield the corresponding vertex partition for the MaxCut solution.

Notes on Implementation

- The interpolation schedule $s(t)$ can be adjusted (linear, quadratic, or sinusoidal) depending on the system size and gap properties.
- The discretized evolution allows the use of standard quantum simulators, where each small time evolution operator $e^{-iH(s_k)\Delta t}$ is applied sequentially.
- The ground state of H_B ensures an easy starting point, and the gradual tuning towards H_P ensures, according to the adiabatic theorem, that the system remains near the instantaneous ground state.